

Exercise – Function – Stochastic representation of the Student's t distribution

This file contains a custom MATLAB function for generating random draws from a Student's t-distribution using its stochastic representation. The function takes the number of draws and the degrees of freedom as inputs and returns the corresponding t-distributed draws.

```
1 function draws = StochasticRepresentation(N,nu)
2 %STOCHASTICREPRESENTATION Generate draws from a Student's t distribution.
3 % Version 1.0, September 2026
4 % Compatible with MATLAB R2014a and later
5 %
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7 %
8 % Description:
9 %   This function generates random draws from a Student's t
10 %   distribution using its stochastic representation. The t-distributed
11 %   random variable is constructed from independent standard normal and
12 %   chi-squared random variables.
13 %
14 % Syntax:
15 %   draws = StochasticRepresentation(N,nu)
16 %
17 % Inputs:
18 %   N - Number of random draws to generate.
19 %   nu - Degrees of freedom. Must be positive.
20 %
21 % Outputs:
22 %   draws - Vector containing t-distributed random draws.
23 %
24 % Notes:
25 %   - The stochastic representation of the Student's t-distribution is
26 %
27 %           Z
28 %   T = -----
29 %          sqrt(U/nu)
30 %
31 %   where
32 %
33 %           Z ~ N(0,1)
34 %
35 %   and
36 %
37 %           U ~ Chi-square(nu).
38 %
39 %   - The random variables Z and U are assumed to be independent.
40 %
41 % ----- BEGIN FUNCTION BODY BELOW -----
42
43 % Generate standard normal random variables
44 Z = random('Normal',0,1,N,1);
45
```

```
46 % Generate chi-squared random variables
47 U = random('Chi-square',nu,N,1);
48
49 % Construct t-distributed random variables
50 draws = Z ./ sqrt(U./nu);
51
52 end
```